

Overview

This page summarizes research-backed design directions for extending TransitSense beyond the bus MVP into **local train and metro** contexts.

The key pattern across the literature is:

1. **Real-time information reduces uncertainty**, but only if passengers trust it.
2. **Route maps strongly shape path choice**, so map design is not cosmetic — it changes rider behavior.
3. **Crowding information is useful even if only some passengers act on it.**
4. **Physical and digital wayfinding must reinforce each other**, especially for first-time users and interchanges.
5. For SIH, the best extension is not full rail operations control — it is a **clear passenger guidance layer** built on top of the bus MVP logic.

Core sources reviewed

1. Heavy-rail real-time arrival signage study (MBTA)
2. Impact of real-time crowding information: a Stockholm metro pilot study
3. Mind the Map! The Impact of Transit Maps on Path Choice in Public Transit
4. Research on Users' Willingness to Use the Urban Subway Wayfinding Signage System: Wuxi Subway
5. Mind the Gap: Navigating the Space Between Digital and Physical Wayfinding in Public Transit

Note: Some sources were available mainly as public summaries or abstract-level pages rather than fully open detailed text. The conclusions below focus on repeated design lessons rather than over-precise technical claims.

What the literature consistently says

1. Rail users need guidance more than raw tracking

In buses, the most important questions are often:

- where is my bus,
- when will it arrive,
- how crowded is it?

In rail systems, those still matter, but passengers also need:

- which line to take,
- which platform or direction is correct,
- where to transfer,
- and how crowded different train cars or interchange paths are.

So the rail extension should emphasize **network guidance and interchange clarity**, not just vehicle tracking.

2. Real-time arrival information helps, but trust is fragile

The heavy-rail countdown-sign literature shows that arrival signage can sharply reduce passengers' overestimation of waiting time. This is good for perceived service quality.

But it also shows a limit: real-time displays are only helpful when they feel credible.

Design implication for TransitSense

Always pair rail ETA or next-train info with:

- clear destination direction,
- platform information,
- and freshness / live status.

3. Crowding information is powerful even with partial use

The Stockholm metro pilot is especially relevant for TransitSense. Even though only a subset of passengers noticed and acted on real-time crowding information, the system still improved passenger distribution across train cars.

Design implication for TransitSense

A metro extension does not need exact every-passenger control. Even a simple display such as:

- **Front coaches crowded**
- **Middle coaches moderate**
- **Rear coaches less crowded**

can improve user decisions and make the system feel advanced.

4. Transit maps strongly influence route choice

The map-design literature shows that passengers often trust schematic maps more than their own intuition. Small layout changes can alter which routes users choose.

Design implication for TransitSense

Route maps must be treated as **decision tools**, not decoration.

This means:

- transfer complexity should be visually honest,
- destinations should be dominant,
- line color consistency matters,
- and overly distorted layouts can mislead first-time users.

5. Digital and physical wayfinding work best together

The wayfinding literature emphasizes that users switch between digital aids and physical surroundings.

Design implication for TransitSense

The extension layer should combine:

- a digital route-map view,
- station / platform / interchange cues,
- and kiosk/signage logic that mirrors the same structure.

This is especially important for first-time users and tourists.

Method / design families

A. Arrival-board extension for rail

Problem

Show passengers when the next train comes and which direction it serves.

Data

- train or timetable ETA
- line ID / destination
- platform / direction metadata
- freshness status

Method

Reuse the bus-kiosk logic, but adapt each row to rail terminology:

- line name/color,
- destination,
- platform,
- minutes to arrival,
- crowd band if available.

Privacy stance

Strong. No personal data required.

Hardware cost level

Low.

Connectivity assumptions

Can work with cached static station data plus live arrival values.

Fit for TransitSense

Very strong reuse candidate.

B. Schematic route-map and interchange guidance

Problem

Help users understand how to traverse a rail network, especially when transferring.

Data

- line graph / station graph
- interchange nodes
- line colors / labels

- destination names

Method

Create a simplified graph-based map rather than an exact geographic map.

Design rules from the literature

- keep a stable color-language,
- minimize unnecessary distortions,
- make transfer nodes prominent,
- show destination direction clearly,
- avoid overloaded labels.

Constraints

- Map design changes passenger behavior, so poor design can mislead.
- Overly complex maps weaken first-time-user confidence.

Privacy stance

Strong.

Hardware cost level

Low.

Connectivity assumptions

Excellent offline, since map assets can be bundled locally.

Fit for TransitSense

Best core extension module.

C. Coach/platform crowding guidance

Problem

Help riders choose less crowded train cars or waiting zones.

Data

- coach-level or approximate train-section crowd status
- platform zone mapping
- train arrival timing

Method

Use a simple left/middle/right or front/middle/rear crowd indicator rather than precise coach telemetry.

Constraints

- Real coach-level sensing is difficult in a hackathon.
- Must avoid false precision.

Privacy stance

Can remain strong if based on aggregate counts or simulated labels.

Hardware cost level

Low to medium depending on how realism is achieved.

Connectivity assumptions

Works well if the crowd labels are simulated or aggregated.

Fit for TransitSense

Strong demonstration feature, but best as a simulated extension.

D. Full interactive trip-planning kiosk

Problem

Support network exploration, transfer planning, and user-specific route decisions.

Data

- network graph
- service state
- interchange logic
- station exits / landmarks

Method

Interactive system with route search and guided transfers.

Constraints

- Rich, but too broad for a first hackathon prototype.
- More UI and data-structure complexity.

Privacy stance

Strong if user queries are not stored.

Hardware cost level

Medium.

Connectivity assumptions

Needs offline fallback screens to remain usable.

Fit for TransitSense

Future enhancement, not first extension.

Comparison table

Design principles for TransitSense

Recommended extension story

Build the train/metro extension around three ideas:

1. Reuse ETA-style passenger information
2. Add graph-based route and interchange guidance
3. Optionally simulate coach crowding guidance

Why this is the right SIH framing

- It reuses the strongest bus MVP modules.

- It adds a clear network-level intelligence story.
- It avoids overcommitting to complex live train integrations.
- It gives judges a believable scale-up path.

Constraints and risks to make explicit

1. Rail route maps can mislead if too distorted.
2. Interchange logic must be simple and clear.
3. Coach-level crowding claims should stay approximate unless well supported.
4. The extension should not depend on complex live railway integrations.
5. Static network guidance may matter more than live train telemetry in the demo.

TransitSense recommendation

Default choice

Extend TransitSense with a **rail arrival board** + **schematic route-map/interchange guidance layer**.

Add if time permits

Add **simple crowding guidance by train section or coach group**.

Avoid for first extension

Do not center the extension on:

- full trip planning,
- exact real-time train operations integration,
- or highly detailed geographic maps.

The best SIH extension is one that feels broader than the bus MVP, while still being explainable and demoable.